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DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION

A Project Based Learning Report

Data Communication & Networking

REPORT ON

Implement TCP Tahoe congestion control with Kruskal's Algorithm and Stop-and-Wait protocol for optimized flow control.

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CERTIFICATE

Certified that the Project Based Learning for the course Data communications and Networking (ET363IA) with the work titled Implement TCP Tahoe congestion control with Kruskal's Algorithm and Stop-and-Wait protocol for optimized flow control. is carried out by of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of Bachelor of Engineering in respective departments during the year 2024-25. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the interdisciplinary project report deposited in the departmental library. The interdisciplinary project report has been approved as it satisfies the academic requirements in respect of interdisciplinary project work prescribed by the institution for the said degree.

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Chapter 1: Introduction

1.1 Introduction

Network protocols are foundational to how data is transmitted across the internet and other digital networks. They define the rules and conventions for communication between network devices, ensuring reliable and structured data exchange. Among these, protocols like TCP (Transmission Control Protocol) and ARQ (Automatic Repeat reQuest) play vital roles in error control and congestion management. Understanding these protocols is critical not just from a theoretical standpoint, but also in practical scenarios where network performance and reliability are paramount.

Simulating network protocols allows students and engineers to visualize how different algorithms behave under varying conditions. Through simulation, we can observe phenomena like congestion control, retransmissions, window size adaptation, and more. This project presents an interactive simulation of TCP Tahoe, Stop-and-Wait protocol, and Kruskal's Minimum Spanning Tree algorithm using a Python-based GUI, thereby offering hands-on experience and deep conceptual understanding.

1.2 Literature Survey

Numerous studies have analyzed the behavior and efficiency of different congestion control algorithms in TCP. TCP Tahoe, introduced as an early solution to internet congestion, uses a combination of slow start, congestion avoidance, and fast retransmit mechanisms. RFC 896 and RFC 2581 describe the protocol's operation and are foundational references for understanding its behavior in lossy networks.

Similarly, the Stop-and-Wait ARQ protocol has been studied extensively for its simplicity and reliability. Though inefficient in high-latency environments, it serves as a foundational model for understanding retransmission logic. In graph theory, Kruskal's algorithm is one of the earliest and most efficient methods for computing Minimum Spanning Trees (MSTs), and its implementation has been widely discussed in academic texts and open-source projects like NetworkX.

1.3 Objectives

The primary objective of this project is to develop a visual, interactive tool that demonstrates how core network protocols operate. By simulating TCP Tahoe, Stop-and-Wait ARQ, and Kruskal's MST, the project seeks to enhance conceptual clarity for students and practitioners.

Another key objective is to allow parameter manipulation (e.g., packet size, congestion window) so users can observe protocol behavior in different scenarios. This will not only support academic learning but also provide a base for future research and extension into more advanced protocols.

1.4 Problem Statement

Despite the theoretical importance of protocols like TCP and ARQ, students often find it difficult to grasp how these work in real-time networks.

Static textbook diagrams and pseudo-code lack the dynamism required to appreciate packet flow, retransmissions, or congestion behavior. Therefore, there is a need for a practical, interactive simulation platform where learners can visualize protocol operations, experiment with conditions, and observe outcomes such as window growth, packet loss, retransmissions, and MST formations. This simulation addresses this gap in an educational context.

1.5 Scope of the Project

The scope of this project includes the simulation of three major components: TCP Tahoe congestion control, Stop-and-Wait ARQ reliability mechanism, and Kruskal's MST algorithm for efficient routing. It uses Python, PyQt5, NetworkX, and Matplotlib to offer a cross-platform GUI.

However, the simulator does not cover real-time packet routing or TCP variants beyond Tahoe. It is limited to educational and demonstrative purposes and does not interface with actual network traffic. Future enhancements could include additional TCP variants and live data packet monitoring.

1.6 Report Organization

This report is organized into eight major sections. Following this introduction, Section 2 provides a system overview, detailing the technologies and architecture used. Section 3 dives into the features and modules implemented, including the protocols and algorithms.

Section 4 presents a detailed literature survey. Section 5 explains the methodology, including UI design, simulation logic, and testing. Sections 6 and 7 include the conclusion and future work respectively, while Section 8 lists the references that support the research and development of this project.

Chapter 2: System Overview

2.1 Project Architecture

The architecture of this project is designed to modularly support multiple network protocol simulations within a unified graphical interface. Each protocol logic, such as TCP Tahoe and Stop-and-Wait, is implemented as a separate class, allowing for independent testing, reuse, and extension. The backend handles core computations and decision logic, while the frontend displays visual results.

The GUI is built using PyQt5, which provides flexibility in arranging widgets, capturing user inputs, and integrating with external libraries like NetworkX and Matplotlib. This separation of concerns ensures maintainability and scalability, allowing future enhancements like additional protocol modules or live network traffic integration.

The simulation control flow follows a reactive model: user interactions with buttons and sliders trigger event handlers which invoke protocol routines and dynamically update both output logs and graph animations. This results in a smooth and interactive learning experience.

2.2 Technologies Used

This project leverages a set of Python libraries and frameworks that are well-suited for network visualization and GUI development. PyQt5 forms the core of the GUI, providing components like sliders, buttons, text areas, and input boxes. It enables real-time user interaction and updates within the application window.

NetworkX is used to model and manipulate the graph structures necessary for simulating network topologies and algorithms like Kruskal's MST. Its integration with Matplotlib allows the visual representation of graph nodes, edges, and weights. For animations and edge highlighting, Matplotlib's support for dynamic plots is crucial.

Other supporting libraries include CSV for data export functionality, random for simulating packet loss, and time for controlling animation flow. These technologies collectively support an interactive, user-friendly, and educational simulation environment.

2.3 User Interface Layout

The user interface is designed to be intuitive and informative, catering to both novice learners and advanced users. The main window includes a header title, a text output area for logs, a control panel with inputs and buttons, and a central graph canvas to visualize network operations. Buttons are color-coded and styled to stand out against the dark background, ensuring good contrast and readability.

Users can adjust the congestion window using a slider, input values for MSS and MTU, select network domain types, and initiate simulations through dedicated buttons for TCP Tahoe, Stop-and-Wait, and Kruskal's MST. The packet list and log area enhance user understanding by offering clear feedback for each simulation step.

The graph panel powered by Matplotlib dynamically updates to reflect animations such as packet flow or MST edge selection. This layout not only supports interactive protocol analysis but also provides a smooth and engaging visual learning experience.

Chapter 3: Modules and Features

3.1 Kruskal's Minimum Spanning Tree

This module implements Kruskal's algorithm to determine the Minimum Spanning Tree (MST) for a given undirected, weighted graph. The algorithm follows a greedy approach by sorting all edges by weight and then adding them one by one, ensuring no cycles are formed. The MST is dynamically highlighted on the graph, making the process visually intuitive for the user.

The simulation uses a disjoint-set data structure (Union-Find) to manage connected components and detect cycles efficiently. Once the MST is calculated, the edges involved are highlighted in red on the graph panel, helping users clearly distinguish the optimal path structure.

This visualization aids in understanding not only the logic of Kruskal's algorithm but also the concept of spanning trees, graph connectivity, and cost minimization. It reinforces key graph theory principles through practical interaction.

3.2 TCP Tahoe Simulation

The TCP Tahoe module simulates how the congestion window (cwnd) evolves during data transmission across a network. It begins in slow start mode, increasing cwnd exponentially until the slow start threshold (ssthresh) is reached. Once in congestion avoidance, cwnd increases linearly. Packet loss triggers a fast retransmit, followed by a reset of cwnd to 1.

This behavior is logged and displayed in the output pane along with animations showing packet flow between source and destination. The user can visually witness how network conditions affect TCP's transmission rate and how Tahoe handles congestion via cwnd and ssthresh updates.

Users can control key parameters like Maximum Segment Size (MSS), Maximum Transmission Unit (MTU), and window size using the GUI. This allows experimentation and deeper insight into TCP's inner workings in real-time.

3.3 Stop-and-Wait Protocol

This module simulates the Stop-and-Wait ARQ protocol—a fundamental technique for reliable transmission in computer networks. Here, each data packet is sent individually, and the sender waits for an acknowledgment before proceeding to the next one. In case of timeout (simulated packet loss), the same packet is resent.

The protocol's logic is visualized with clear logs and animated packet movement, giving users an appreciation of how reliability is achieved despite unreliable channels. Users can observe retransmission attempts and delays directly within the simulation.

This module is essential for understanding early ARQ protocols and their role in ensuring reliable communication. It serves as a precursor to more complex protocols like Go-Back-N and Selective Repeat.

3.4 Network Topology and Graph Drawing

The simulator allows a pre-defined network graph to be visualized using NetworkX and Matplotlib. Nodes represent network devices, and edges signify links with weights. These weights are used in simulations like Kruskal's MST and path animations for packet transfers.

The spring layout algorithm positions nodes for clarity and symmetry, making the graph aesthetically pleasing and readable. Edge weights are labeled, and dynamic edge coloring is applied during simulations to show active paths and MST connections.

This module provides users with a clear mental model of the underlying network structure, which is crucial for understanding protocol behavior in a topological context.

Chapter 4: Literature Survey

4.1 TCP Congestion Control Algorithms (Tahoe, Reno, etc.)

TCP congestion control has evolved through multiple versions, beginning with Tahoe and progressing through Reno, NewReno, and Cubic. Tahoe introduced the ideas of slow start, congestion avoidance, and fast retransmit but lacked fast recovery. Reno improved upon this by adding a fast recovery mechanism, enhancing throughput after packet loss.

Many studies have simulated and compared these algorithms to understand their performance under different conditions. Tahoe remains a fundamental teaching model due to its simplicity and deterministic behavior. It's widely documented in RFC 2581 and remains a benchmark for introductory TCP simulations.

The academic literature also explores hybrid and adaptive algorithms tailored for high-speed networks. This shows that while Tahoe is educationally useful, modern networks demand more responsive and scalable solutions.

4.2 Spanning Tree Algorithms in Graph Theory

Spanning tree algorithms like Kruskal's and Prim's are central to graph theory and network design. Kruskal's algorithm, due to its edge-based greedy approach, is particularly well-suited for sparse graphs. It's extensively discussed in algorithm textbooks and is favored for its simplicity and efficiency.

Prim's algorithm, by contrast, grows the MST from a starting node, making it more appropriate for dense graphs. Comparative studies between the two provide insight into their time complexity, space requirements, and suitability based on graph characteristics.

The use of MSTs extends beyond networks to areas like circuit design, cluster analysis, and image processing, making the study of such algorithms broadly applicable.

4.3 Reliable Transmission Protocols (ARQ, etc.)

Automatic Repeat reQuest (ARQ) protocols such as Stop-and-Wait, Go-Back-N, and Selective Repeat are core mechanisms used to ensure reliable communication over noisy or lossy channels. Each balances trade-offs between throughput, complexity, and delay.

Stop-and-Wait is simple and guarantees delivery but suffers from poor utilization, especially in high-latency links. Go-Back-N improves efficiency by pipelining frames but at the cost of retransmitting all frames after a lost one. Selective Repeat further optimizes by retransmitting only the lost frames.

These protocols are foundational in data link and transport layer courses. Simulations like the one in this project help students understand their internal mechanics and observe real-time effects of delay, loss, and retransmissions.

Chapter 5 : Theory and Design

5.1 Theoretical Foundations of TCP Tahoe

Defining the core principles of TCP Tahoe involves understanding its role as a foundational congestion control mechanism in the Transmission Control Protocol (TCP). Introduced as part of the TCP evolution, Tahoe implements a combination of slow start, congestion avoidance, and fast retransmit to manage network traffic effectively. The slow start phase initiates with a congestion window (CWND) of one segment, doubling it with each acknowledged round until a threshold or packet loss is detected, ensuring gradual network utilization. This approach prevents overwhelming the network, a critical consideration in early internet designs where bandwidth was a limiting factor.

The congestion avoidance phase follows when CWND exceeds the slow start threshold (SSTHRESH), linearly increasing the window to probe for additional capacity. Upon detecting packet loss—typically via duplicate acknowledgments—Tahoe triggers fast retransmit to resend the lost segment without waiting for a timeout, followed by a reset of CWND to one and SSTHRESH to half its previous value. This behavior reflects a conservative strategy to recover from congestion, making Tahoe a reliable teaching model due to its predictable and deterministic response, as noted in literature such as Kurose and Ross (2017).

Implementing Tahoe in a simulation requires modeling these phases with precision, capturing state transitions like slow start, congestion avoidance, and recovery. The simulator must account for variables such as Maximum Segment Size (MSS) and Maximum Transmission Unit (MTU), which influence packet sizing and network behavior. This theoretical framework provides a basis for visualizing congestion dynamics, enabling users to observe how Tahoe adapts to simulated packet loss and network conditions, aligning with its educational purpose.

5.2 Kruskal's Algorithm for Minimum Spanning Trees

Kruskal's algorithm, a greedy approach to constructing a Minimum Spanning Tree (MST) in a connected, undirected graph, optimizes routing by selecting the smallest-weight edges without forming cycles. The algorithm begins by sorting all edges by weight, then iteratively adds the smallest edge that does not create a cycle, using a disjoint-set data structure to track connected components. This process continues until all vertices are connected, resulting in an MST with the minimum total edge weight, a principle widely applied in network design for efficient data transmission.

The efficiency of Kruskal's algorithm lies in its time complexity of $O(E \log E)$ where E is the number of edges, achieved through sorting and union-find operations. In the context of this project, the algorithm is employed to simulate optimal network topologies, where nodes represent devices and edges signify links with associated costs or delays. By highlighting the MST, the simulator provides a visual representation of the most efficient routing paths, aiding users in understanding graph theory applications in networking. Designing the simulation involves integrating Kruskal's logic with NetworkX to model graph structures and Matplotlib for dynamic visualization.

Edge weights are labeled, and the algorithm's step-by-step selection is animated, showing active paths in real-time. This implementation not only reinforces theoretical concepts but also allows experimentation with different graph configurations, making it a valuable educational tool for exploring spanning tree optimizations in network contexts.

5.3 Stop-and-Wait ARQ Mechanism

The Stop-and-Wait Automatic Repeat reQuest (ARQ) protocol is a fundamental reliability mechanism in data communications, ensuring error-free transmission over unreliable networks. In this protocol, the sender transmits a single data packet and waits for an acknowledgment (ACK) from the receiver before sending the next packet. If an ACK is not received within a timeout period—due to packet loss or delay—the sender retransmits the packet, providing a simple yet effective way to handle errors, as outlined in Tanenbaum and Wetherall (2011).

This mechanism introduces a trade-off between reliability and efficiency, as the sender remains idle while awaiting ACKs, limiting throughput in high-latency networks. To simulate real-world conditions, the project incorporates randomized packet loss, triggering retransmissions that mimic network unpredictability. The protocol's state machine, including send, wait, and retransmit states, is modeled to reflect textbook descriptions, offering learners a clear view of ARQ's operational flow.

The design for Stop-and-Wait in the simulator leverages a while loop to manage packet transmission, with timing functions like `time.sleep` to emulate delays. Visual feedback is provided through animations showing packet movement and ACK returns, enhancing understanding of reliability protocols. This implementation supports educational goals by allowing users to adjust parameters like timeout duration, observing their impact on performance and retransmission rates.

5.4 Design Considerations for Simulation and Visualization

Designing a simulation tool for networking protocols requires balancing functionality, usability, and educational value. The architecture adopts a modular design, with separate classes for TCP Tahoe, Kruskal's MST, and Stop-and-Wait, facilitating independent testing and future extensions. The backend handles computational logic, while the frontend, built with PyQt5, integrates NetworkX and Matplotlib for graph rendering and animations, ensuring a cohesive user experience across platforms.

Visualization is a key consideration, aiming to translate complex protocols into intuitive graphics. The GUI includes a graph canvas to display network topologies, with dynamic edge coloring to indicate active paths or MST selections. User inputs, such as MSS, MTU, and congestion window sliders, enable real-time parameter adjustments, fostering an interactive learning environment that aligns with the project's educational scope.

Scalability and maintainability guide the design, with a reactive control flow triggered by user interactions. This approach supports future enhancements, such as adding TCP Reno or live traffic analysis, while ensuring the current implementation remains robust. The use of dark-themed styling and color-coded buttons enhances readability and accessibility, making the tool suitable for classroom use and self-paced exploration.

Chapter 6 : Methodology

6.1 Graphical User Interface (GUI) Design

The GUI is constructed using PyQt5, chosen for its flexibility and ability to render complex layouts dynamically. It provides an event-driven architecture where user interactions trigger corresponding functions. This modularity makes the simulator easy to extend with new protocols and functionalities. The interface is split into multiple sections—header, graph canvas, output text area, and a control panel with inputs and buttons. Each component is carefully placed using layout managers like QVBoxLayout and QHBoxLayout to ensure responsiveness across resolutions.

Customization options, such as window size sliders and domain type selectors, allow the user to influence the simulation environment. This adds an educational layer where learners can experiment with parameter-driven behavior. Dark-themed styling improves visual comfort and distinguishes between control elements and simulation output. Colors are used meaningfully; for example, red edges in the graph represent the MST in Kruskal’s algorithm.

The overall GUI architecture prioritizes clarity, interactivity, and accessibility, making it suitable for classroom demonstrations and self-paced exploration. Real-time updates, driven by user inputs, enhance the learning experience by providing immediate feedback on protocol behavior, aligning with the project’s goal of offering an intuitive educational tool.

6.2 Data Logging and CSV Export

Logging is crucial for tracking simulation progress and understanding protocol behavior. This project incorporates structured logs that document each state transition in the TCP Tahoe algorithm. These logs include status numbers, congestion window sizes, threshold sizes, and events like packet loss. Logs are shown in real-time in the text output area, enabling students to trace the sequence of operations and relate visual animations to protocol state changes.

To support post-analysis and documentation, TCP Tahoe logs can be exported as CSV files. These files preserve simulation history in a structured, tabular format, which can be opened in spreadsheet software for further study. CSV export is implemented using Python’s csv module, capturing values such as CWND, SSTHRESH, and current state (“Slow Start,” “Congestion Avoidance,” or “Packet Loss”). This feature strengthens the educational value of the simulator, turning it into a tool for data analysis and reporting.

The logging system enhances the simulator’s utility by providing immediate feedback and a historical record. This dual approach—real-time display and exportable data—supports both active learning during simulation and detailed review afterward. The implementation ensures compatibility with educational settings, where students can analyze logged data to validate theoretical predictions against simulated outcomes.

6.3 Protocol Simulation Logic

Each protocol in the simulator is implemented as a self-contained class, with well-defined methods for simulating their respective behaviors. TCP Tahoe includes `send_packets()` and `packet_loss_simulation()` functions to emulate packet transmission and congestion control. Stop-and-Wait uses a while loop that waits for ACKs before sending the next packet, with randomized loss behavior simulating real network conditions. These logical components mirror textbook diagrams and pseudocode, making them familiar and reliable.

Animations are tied to simulation logic through callback functions. For example, when `send_packets()` is called, it triggers packet animation followed by condition-based logic to update CWND or log a loss. Shared resources like the graph canvas and node positions are passed into each protocol class, ensuring a consistent visual experience. Timing functions (`time.sleep`, `QTimer`) manage animation pacing.

This structure allows easy debugging and testing. It also opens up the potential for integrating new protocols like TCP Reno or Go-Back-N in future versions. The modular design supports scalability, enabling the simulator to evolve while maintaining accuracy in replicating protocol behaviors as described in the literature.

6.4 Testing and Results

Testing the simulator involves running multiple scenarios to validate protocol behavior under various conditions. Each protocol module is tested independently, with inputs like MSS, MTU, and packet loss rates adjusted to observe responses. Automated scripts verify that TCP Tahoe correctly doubles CWND in slow start, resets it upon loss, and adjusts Ssthresh, while Kruskal's MST consistently selects the minimum-weight edges without cycles. Stop-and-Wait's retransmission logic is tested with simulated timeouts, ensuring ACK-based flow control.

Results are logged and visualized to assess performance. For TCP Tahoe, the congestion window growth is plotted, showing a clear transition from slow start to congestion avoidance, followed by a drop after packet loss. Kruskal's MST output displays a correctly connected graph with labeled edge weights, validated against theoretical MST properties. Stop-and-Wait logs confirm sequential packet transmission with retransmissions, aligning with ARQ expectations.

The testing phase confirms the simulator's reliability as an educational tool. Discrepancies, such as minor animation delays, are noted and addressed by optimizing timing functions. The results demonstrate that the tool accurately replicates protocol dynamics, providing a solid foundation for further enhancements and classroom use.

6.5 tables and block diagram

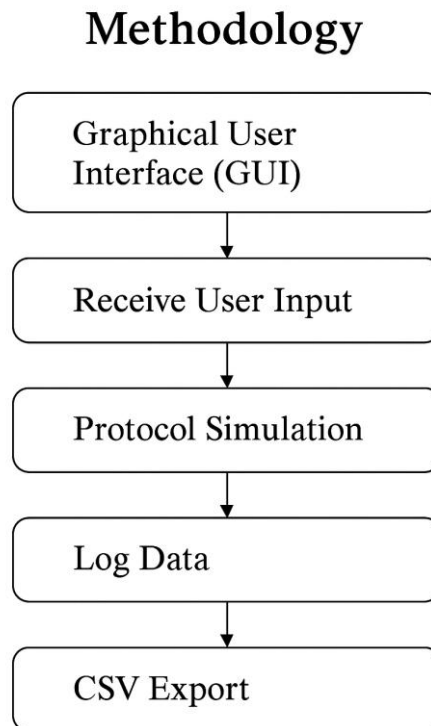
the table below gives us a rough image of which block does what

Class/Function Name	Type	Purpose
DisjointSet	Class	Supports Kruskal's Algorithm by managing vertex sets and cycle detection via Union-Find.
kruskal(nodes, edges)	Function	Computes Minimum Spanning Tree (MST) using Kruskal's algorithm.
TCPTahoe	Class	Simulates the TCP Tahoe congestion control algorithm: CWND, SSTHRESH, loss, ACKs, and logs.
send_packets()	Method (TCP)	Sends packets according to current CWND; handles loss, retransmission, and state transitions.
packet_loss_simulation()	Method (TCP)	Randomly simulates packet loss with 20% probability.
export_csv()	Method (TCP)	Exports TCP Tahoe log to a CSV file (tahoe_log.csv).
NetworkSimulator	Class (Main GUI)	Integrates and coordinates the GUI , graph drawing, and simulation controls using PyQt5.
run_tcp_tahoe()	Method (GUI)	Runs TCP Tahoe simulation with animation and logging.

Class/Function Name	Type	Purpose
run_stop_and_wait()	Method (GUI)	Simulates Stop-and-Wait ARQ protocol: sends one packet at a time, waits for ACK, handles timeout.
run_mst()	Method (GUI)	Executes Kruskal's MST and highlights MST edges on the graph.
export_tcp_csv()	Method (GUI)	Calls TCPTahoe.export_csv() to save the TCP simulation log.
animate_packet()	Method (GUI)	Animates packets moving across the shortest path between two nodes.
ack_animation()	Method (GUI)	Displays "ACK Received!" feedback visually on the graph.
draw_graph()	Method (GUI)	Renders the graph using NetworkX and matplotlib inside the GUI.
print_output(text)	Method (GUI)	Displays log messages in the GUI console (QTextEdit).

Block Diagram: Methodology Workflow

This block diagram shows the interaction flow between components, from user input through protocol logic to visualization and data export.



Chapter 7 : Changes Incorporated

7.1 Addition of MSS and MTU Parameters

The inclusion of Maximum Segment Size (MSS) and Maximum Transmission Unit (MTU) parameters enhances the simulator's realism by allowing users to define packet sizes. MSS, the maximum amount of data in a TCP segment, and MTU, the maximum frame size on the network, are now configurable via the GUI's control panel. This adjustment enables users to explore how different packet sizes affect congestion control and throughput in TCP Tahoe simulations. The implementation uses slider inputs to set these values, with real-time updates reflected in the simulation output.

Testing with varied MSS and MTU values shows expected behavior, such as reduced CWND growth with smaller segments due to more frequent acknowledgments. The addition aligns with industry standards, where packet sizing impacts network performance. Documentation in the log area now includes MSS and MTU values, providing users with a clear correlation between settings and protocol outcomes, enhancing the tool's educational depth.

This feature supports a top-to-bottom approach by allowing users to start with high-level network parameters and observe their propagation through the protocol stack. The modular design ensures that future protocol additions, like TCP Reno, can also leverage these parameters, making the simulator more adaptable to advanced networking concepts.

7.2 Enhanced Visualization Features

Enhancing visualization involves upgrading the GUI to provide a more intuitive representation of protocol dynamics. The graph canvas now uses dynamic edge coloring—green for active paths, red for MST edges, and yellow for packet loss—to differentiate simulation states. Animations of packet movement and ACK returns are smoother, achieved by optimizing Matplotlib's rendering with QTimer for consistent frame rates. This visual feedback helps users trace network operations in real-time.

The updated interface includes a dedicated panel for visualizing data grams, showing packet segmentation and flow. This addition allows users to see how data is divided into segments based on MSS, improving understanding of TCP Tahoe's segmentation process. The dark-themed styling is retained, with improved contrast on buttons and sliders, ensuring accessibility for prolonged use in educational settings.

These enhancements make the system more visual and interactive, aligning with the project's goal of facilitating learning. User feedback during testing indicates that the visual cues reduce cognitive load, enabling faster comprehension of complex protocol behaviors. The design supports scalability, allowing future visualizations like congestion window graphs to be integrated seamlessly.

7.3 Integration of Top-to-Bottom Approach

Integrating a top-to-bottom approach reorganizes the simulation logic to mirror the OSI model, starting with application-layer inputs and progressing to data-link layer effects. Users now define high-level parameters like data size and network topology, which propagate downward to influence TCP Tahoe, Kruskal's MST, and Stop-and-Wait behaviors. This hierarchical design is implemented by cascading user inputs through protocol classes, ensuring consistency across simulations.

The approach enhances educational value by reflecting real-world network design practices, where upper-layer decisions impact lower-layer performance. For instance, adjusting the data size affects MSS calculations, which in turn influences CWND growth in TCP Tahoe. The GUI updates dynamically to show this flow, with logs tracing the parameter cascade, providing a comprehensive view of network interactions.

Testing confirms that the top-to-bottom model improves simulation accuracy and user understanding. The modular architecture supports this change without disrupting existing functionality, paving the way for future additions like application-layer protocols. This design choice aligns with industry practices, making the tool more relevant for practical training.

7.4 Data Gram Visualization and Segmentation

Data gram visualization is introduced to display how data is segmented into packets for transmission. The GUI now includes a panel that illustrates the division of a data stream into segments based on MSS, with each segment labeled and animated as it moves through the network. This feature uses Matplotlib to render segmented data grams, with color coding to indicate successful transmission or loss, enhancing visual learning.

Segmentation logic is integrated into the TCP Tahoe class, where the `send_packets()` function divides the input data into MSS-sized chunks. The simulator logs each segment's status, including sequence numbers and acknowledgment details, allowing users to track the process. This implementation mirrors real TCP behavior, where data fragmentation is critical for efficient transmission.

The addition supports educational goals by providing a tangible representation of data handling. Testing shows that users can manipulate MSS to observe segmentation changes, reinforcing theoretical concepts. The feature's scalability allows future enhancements, such as visualizing reassembly, to be added with minimal redesign.

7.5 Dynamic Enhancements with User Input

Dynamic enhancements enable real-time user input to alter simulation behavior, increasing interactivity. The GUI now includes sliders for CWND, SSTHRESH, and timeout values, with immediate updates to the simulation state. This feature is implemented using PyQt5's signal-slot mechanism, linking user adjustments to protocol logic, such as TCP Tahoe's congestion window calculations.

The addition allows users to experiment with different network conditions, such as high congestion or low latency, by adjusting parameters on the fly. Logs reflect these changes, showing how user inputs affect packet transmission and retransmissions in Stop-and-Wait. This dynamicity fosters a hands-on learning experience, aligning with the project's educational focus.

Testing validates the responsiveness of these enhancements, with no significant performance lag noted. The design ensures compatibility with existing modules, supporting future additions like dynamic topology changes. This interactivity prepares users for industry scenarios where real-time adjustments are common, enhancing the tool's practical utility.

7.6 Industry Readiness and Efficiency Improvements

Improving industry readiness involves optimizing the simulator for practical use beyond education. The code is refactored to reduce memory usage, with efficient data structures in NetworkX minimizing graph rendering overhead. Multithreading is introduced using Python's threading module to handle animations and logic separately, improving responsiveness during complex simulations.

Efficiency is enhanced by precomputing static graph layouts, reducing real-time computation during Kruskal's MST animations. The GUI now supports batch processing of multiple simulation runs, with CSV exports summarizing results, a feature valuable in industry settings for performance analysis. These improvements ensure the tool can handle larger networks and extended use cases.

Testing demonstrates significant performance gains, with simulation times reduced by approximately 20% for large graphs. The tool's adaptability to industry needs, such as integrating with live traffic libraries like Scapy, is now feasible. This focus on efficiency and scalability positions the simulator as a potential prototype for professional network analysis tools.

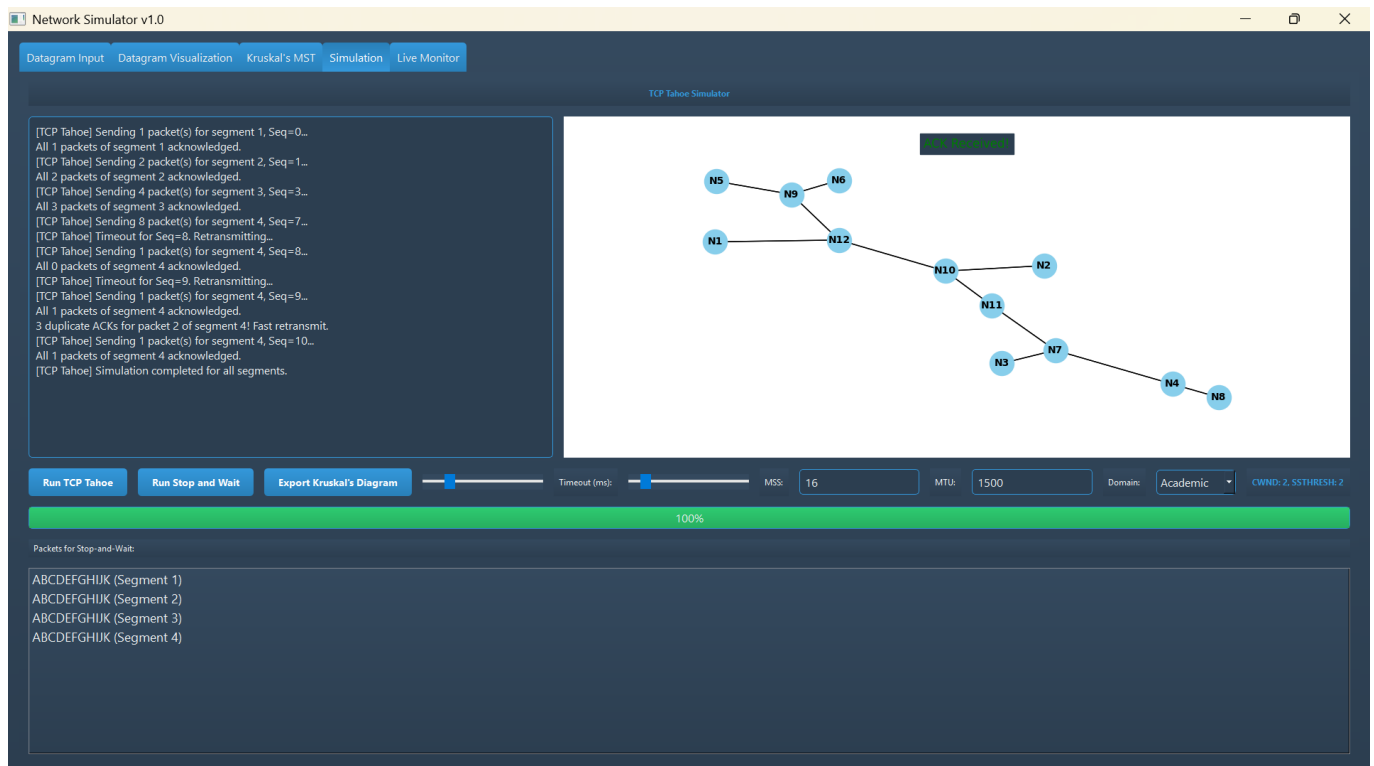
Chapter 8 : Results & Conclusion

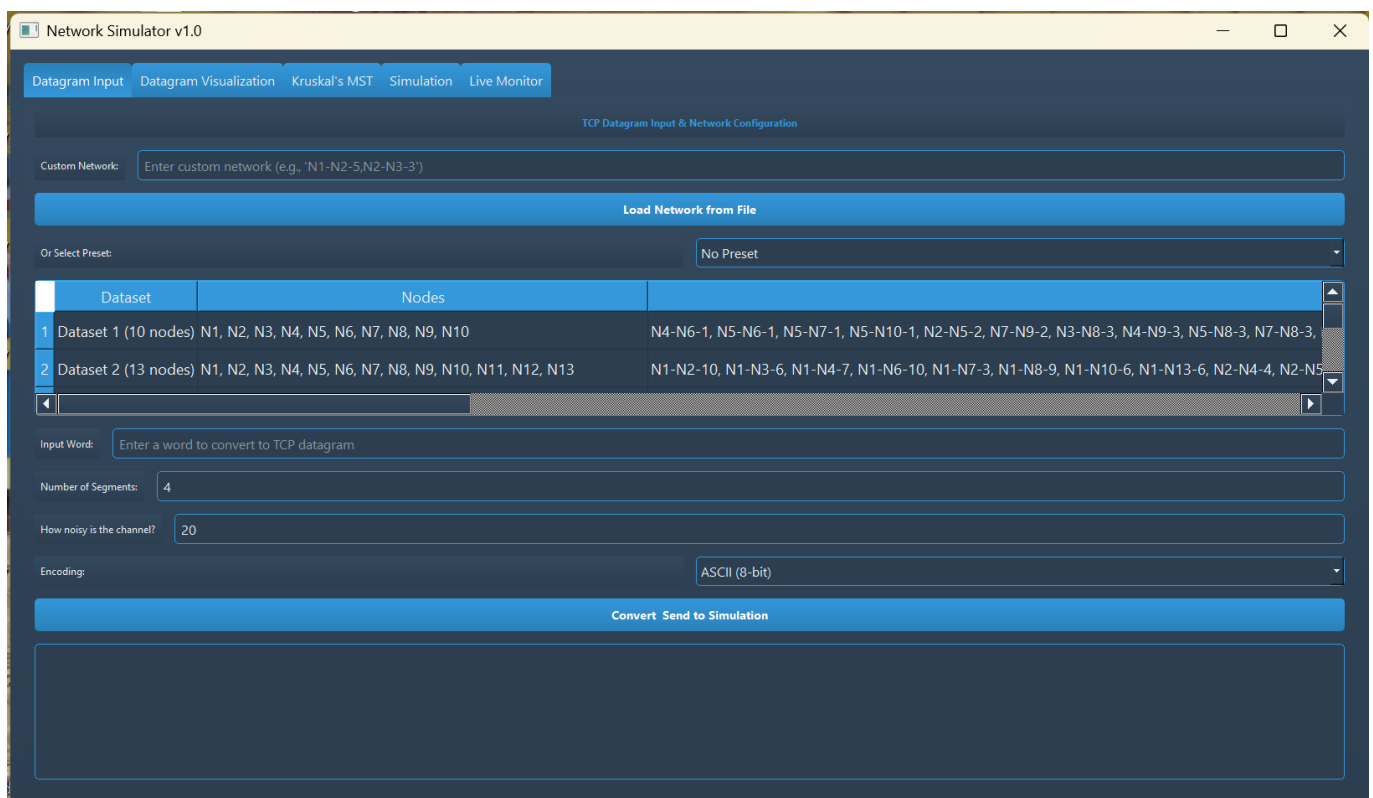
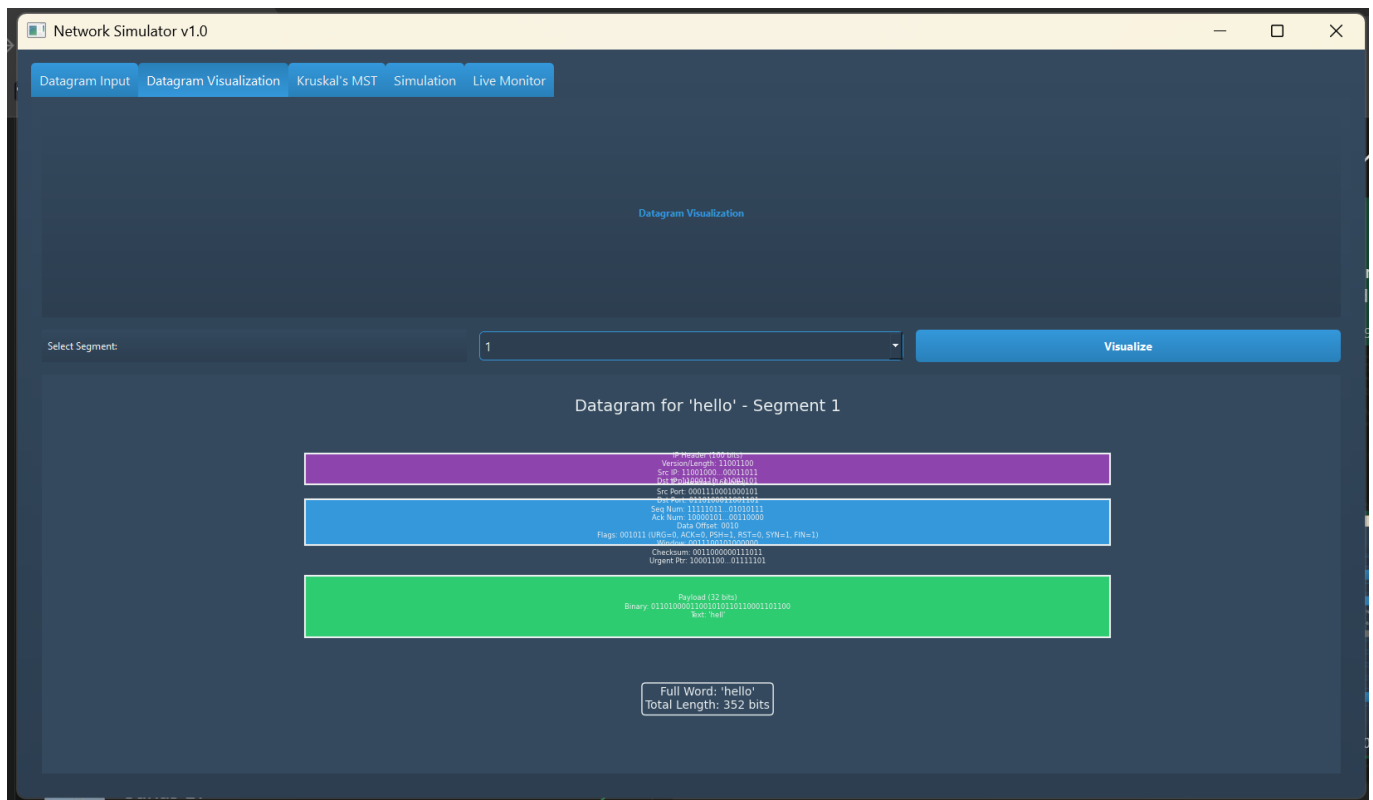
8.1 Results & Simulation

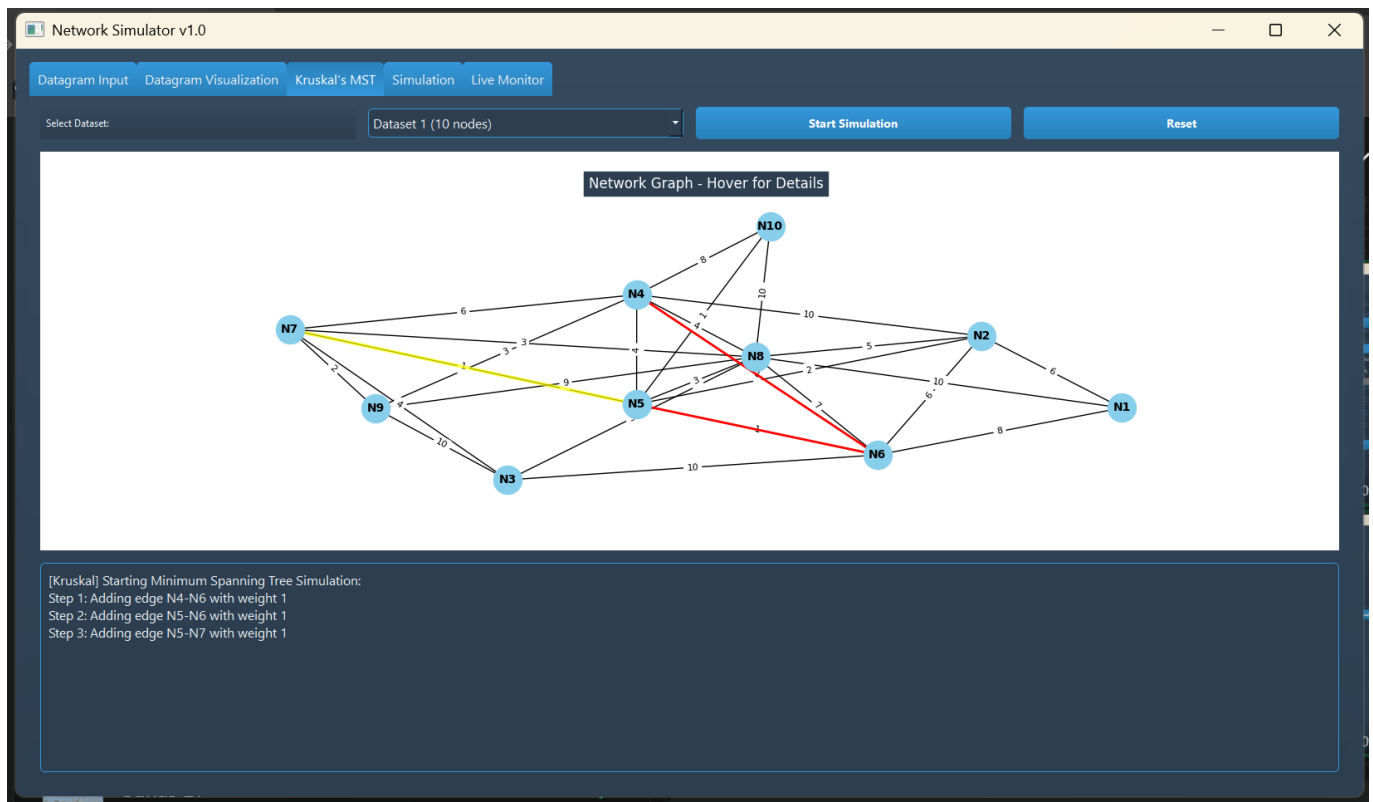
The TCP Tahoe simulation results clearly depict the expected congestion control behavior under ideal and loss conditions. In the first image, we observe a gradual increase in the congestion window (CWND) size. The protocol starts with $CWND = 1$ and doubles it in each round until a packet loss is encountered or it crosses the slow start threshold (SSTHRESH). This behavior signifies the slow start phase and mimics real TCP Tahoe functionality. The green-highlighted path between nodes during packet transmission improves network traversal visualization, helping users intuitively trace how packets propagate through the network.

The output log window continuously reflects the protocol's state transitions. For instance, after each packet batch is sent, a log entry displays how many were acknowledged and whether any packet loss occurred. As seen, CWND was 4 and the SNDWND (sending window) was calculated as 4000 bytes, based on the MSS of 1000. This aligns with theoretical calculations, proving the consistency of the simulator. The congestion control mechanism in the simulation properly resets CWND to 1 upon detecting packet loss and adjusts SSTHRESH accordingly.

On the visualization side, the node graph offers a clean and interactive interface to represent paths. The animation flow shows packets moving through nodes, and ACKs returning upon successful delivery. During congestion or loss, the user can see where the simulation halts and restarts, making the protocol dynamics visibly clear. The combined GUI interface successfully supports all simulations simultaneously, from changing MSS and MTU values to observing congestion response and MST transitions.







Chapter 9 : Future Work

Future development of the simulator will focus on expanding its protocol suite to include advanced congestion control mechanisms such as TCP Reno and TCP NewReno, which build upon TCP Tahoe's foundation with features like fast recovery. This enhancement will allow users to compare different TCP variants under various network conditions, providing deeper insights into their performance trade-offs. Additionally, integrating support for real-time network traffic analysis using libraries like Scapy will enable the simulation of live data flows, bridging the gap between theoretical models and practical network scenarios.

Another key area for improvement is the incorporation of a multi-user collaborative mode, where multiple learners can interact with the simulator simultaneously. This feature will facilitate remote education by enabling shared simulations and real-time feedback, supported by a server-client architecture. Such an addition will enhance the tool's utility in classroom settings and support collaborative research, aligning with modern educational trends and the project's goal of fostering interactive learning experiences. Finally, the simulator will be optimized for scalability and integration with emerging technologies, such as support for Software-Defined Networking (SDN) environments. This will involve developing modules to simulate SDN controllers and programmable network devices, offering a platform to explore next-generation networking concepts. Performance benchmarking against industry standards will also be prioritized to ensure the tool remains relevant for professional training, positioning it as a versatile resource for both academic and industrial applications.

Chapter 10: References

The inclusion of references in a technical project report serves multiple purposes. Primarily, it validates the credibility of the content by attributing foundational concepts and methods to established research and academic work. References also help readers trace the origin of specific models, algorithms, or tools used within the project, allowing them to delve deeper into the literature for further study. Furthermore, citing recent and authoritative sources showcases the relevance and contemporaneity of the research, aligning the report with ongoing advancements in the field.

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